

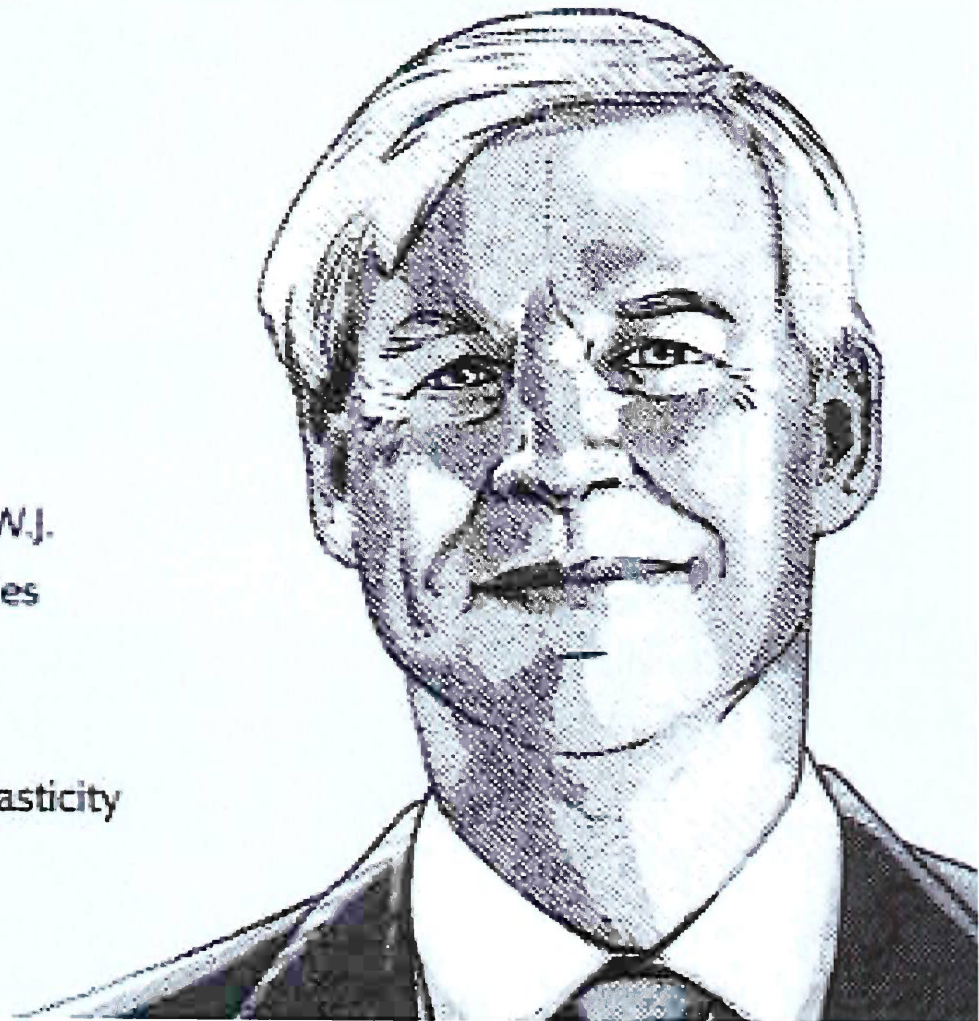
2003 Nobel Prize, Robert F. Engle (Week 8)

Robert F. Engle III

Born: November 10, 1942

Econometrician

- 2003 Nobel Prize Recipient in Economic Sciences, along with Clive W.J. Granger, for their analysis of time-series data with time-varying volatility
- Best known for his development of autoregressive conditional heteroskedasticity (ARCH) modeling and testing



Engle ARCH models (Week 8)

2003 Nobel Prize ARCH model:

$$\sigma_t^2 = \omega + \alpha_1 r_{t-1}^2, \quad \omega > 0, \alpha_1 \geq 0$$

Handwritten annotations for the equation $\sigma_t^2 = \omega + \alpha_1 r_{t-1}^2$:

- t → "today"
- r → steep slope
- α_1 → slope
- $t-1$ → "t-1"

Equity Curve

To directly compare the investment performance of two or more assets, plot the equity curve.

The equity curve shows how a one-dollar investment amount in each asset grows over time.

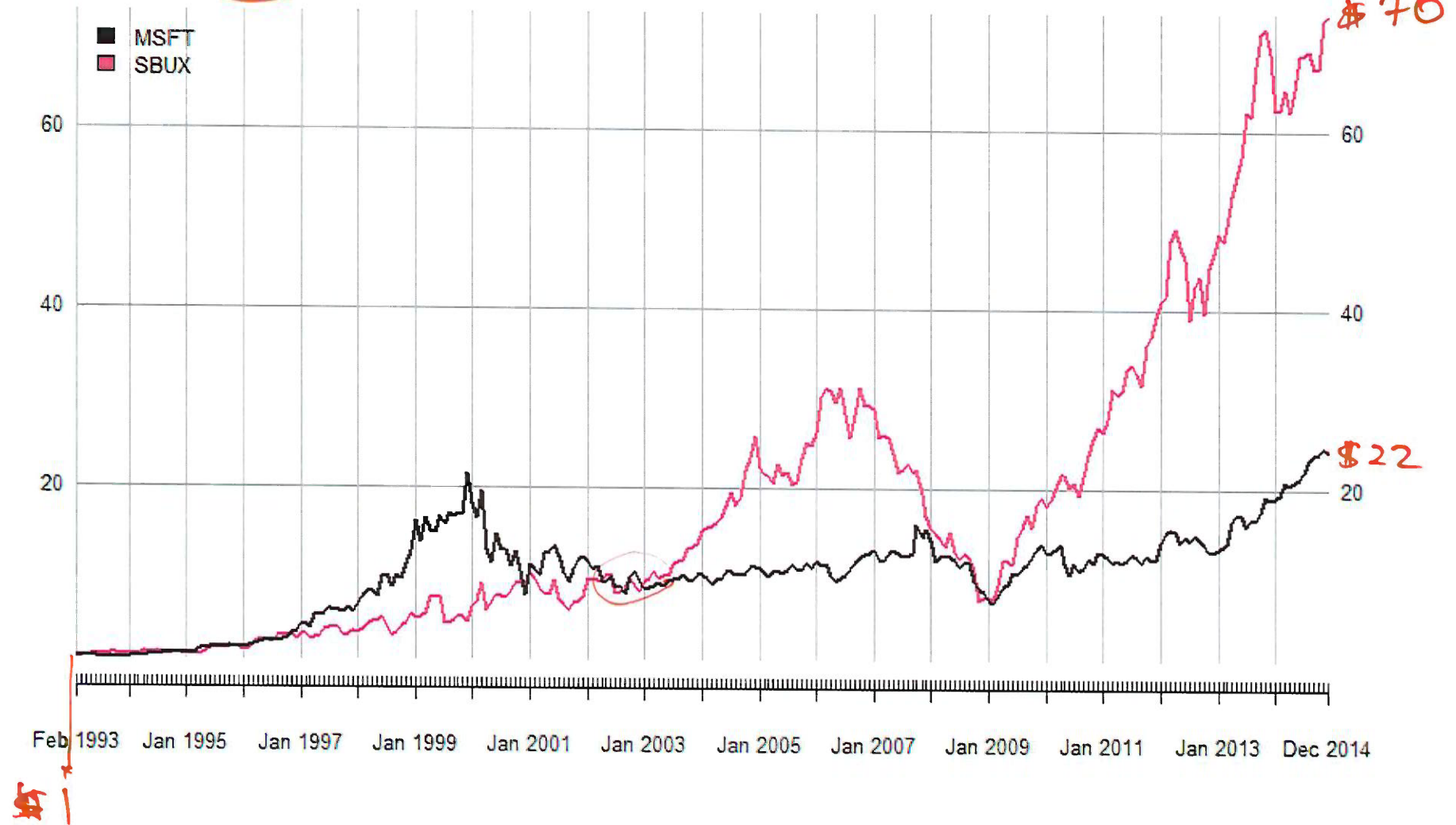
Better-performing assets have higher equity curves.

For simple returns, the k -period equity curve is

$$V_t^{(k)} = \prod_{j=0}^{k-1} (1 + R_{t-j}) .$$

↓
Monthly Equity Curves

Feb 1993 / Dec 2014



Equity Curve

A one-dollar investment in Starbucks dominated a one-dollar investment in Microsoft over the given period. In particular, \$1 invested in Microsoft grew to only about \$22 (over about 20 years) whereas \$1 invested in Starbucks grew to over \$70.

Notice the huge increases and decreases in value of Microsoft during the dot-com bubble and bust over the period 1998–2001, and the enormous increase in value of Starbucks from 2009–2014.

Equity Curve

$K=1$
 π
 $j=0$



NOT
log-return.



Simple-return



```
# equity curves
equityCurveMsft = cumprod(1 + msftMonthly.S>Returns)
equityCurveSbux = cumprod(1 + sbuxMonthly.S>Returns)

dataToPlot = merge(equityCurveMsft, equityCurveSbux)
plot(dataToPlot, main="Monthly Equity Curves",
      legend.loc="topleft")
```

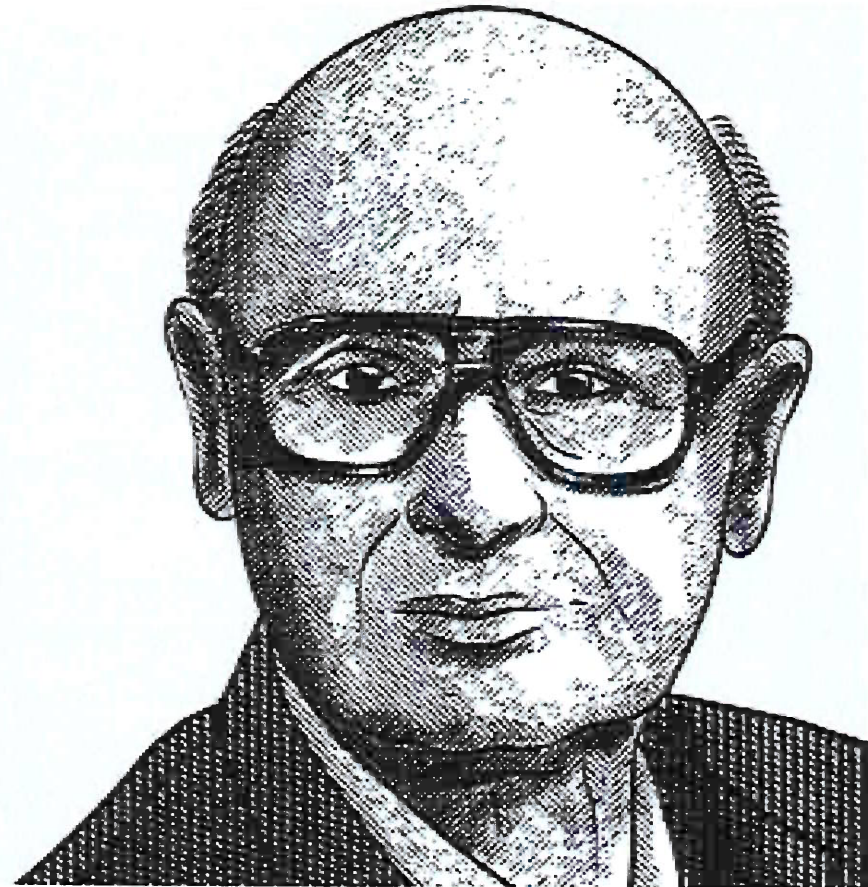

1990 Nobel Prize, Harry Markowitz (Week 4)

Harry Markowitz

Born: August 24, 1927

Economist

- 1990 Nobel Prize Recipient in Economic Sciences for developing the modern portfolio theory
- His work popularized concepts like diversification and overall portfolio risk and return, shifting the focus away from the performance of individual stocks

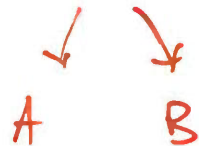


Portfolio (Week 4)

A two-asset portfolio is a mix of two assets where you choose the weights (the proportions of your total investment) allocated to each asset.

For example, an equal-weight two-asset portfolio just means you split your money 50–50.

You choose the weights w to minimise the portfolio variance, and the resulting weights are determined only by the variances and covariances of the assets.



Warren Buffett, Former CEO, Berkshire Hathaway



Warren Edward Buffett is an American investor and philanthropist who is the chairman and former CEO of the conglomerate Berkshire Hathaway. As a result of his success, Buffett is one of the best-known investors in America. [Wikipedia](#)

Born: 30 August 1930 (age 95 years), [Omaha, Nebraska, United States](#)

Net worth: 148.6 billion USD (2026) [Forbes](#)



Buffett's Bet

The table (next slide) shows the simple returns for eight years of a ten-year bet that Warren Buffett placed with Protégé Partners, a New York hedge fund.

Buffett claimed that a simple index fund would **beat** the hedge fund portfolio selected by Protégé over a ten-year timeframe.

At Buffett's shareholder meeting, he provided an update on the current state of the bet.

		X_i	Y_i	$D_i = X_i - Y_i$
	Year	S&P Index	Hedge Funds	
1	2008	-37.0%	-23.9%	-13.1
2	2009	26.6%	15.9%	10.7
	2010	15.1%	8.5%	
	2011	2.1%	-1.9%	
	2012	16.0%	6.5%	
	2013	32.3%	11.8%	
	2014	13.6%	5.6%	
8	2015	1.4%	1.7%	-0.3

Simple return

Use a paired t-test to assess the statistical significance between the two return strategies.

$$\bar{D} = \frac{-13.1 + \dots + -0.3}{8}$$

$$= 5.74$$

percentage points.

$$S_D = \sqrt{\frac{1}{n-1} \left((-13.1 - 5.74)^2 + \dots \right)}$$

$$= 9.69$$

Paired t-test

Define paired differences $D_i = X_i - Y_i$ for $i = 1, \dots, n$, and let $\mu_D = \mathbb{E}[D_i]$ denote the population mean difference.

We test

$$H_0 : \mu_D = 0$$

$$H_1 : \mu_D \neq 0. \quad \text{I want to reject } H_0$$

Compute the sample statistics

$$\bar{D} = \frac{1}{n} \sum_{i=1}^n D_i, \quad s_D = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (D_i - \bar{D})^2},$$

The test statistic

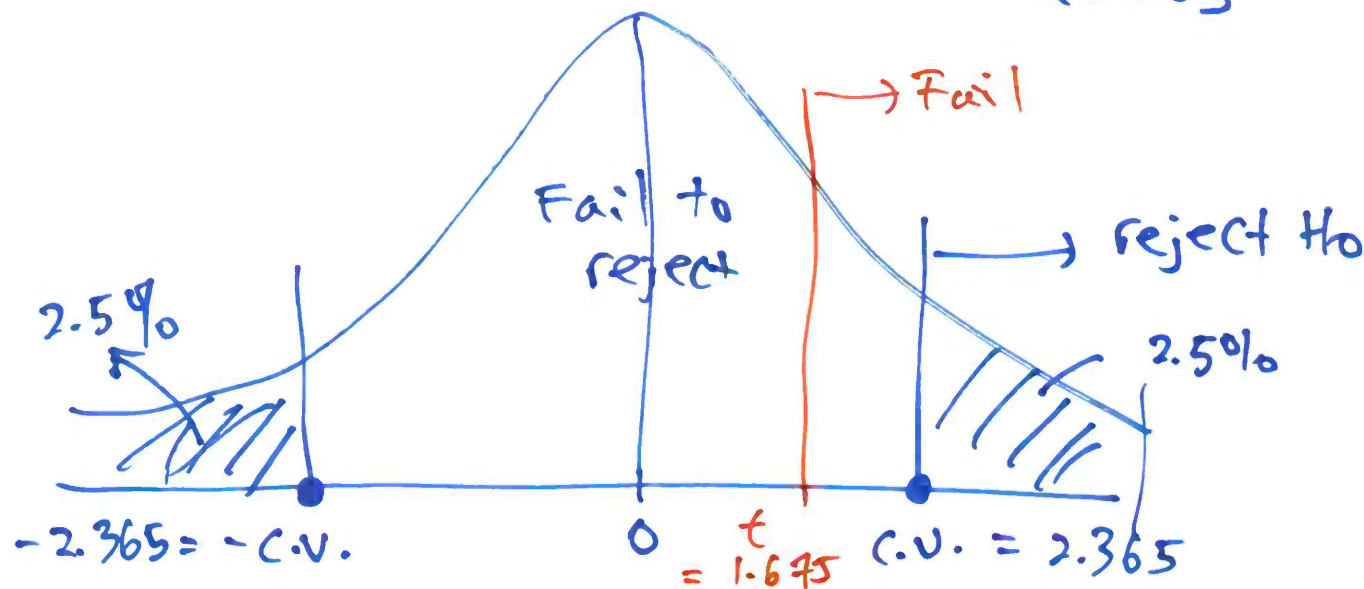
$$t = \frac{\bar{D}}{s_D / \sqrt{n}}, \quad \text{df} = n - 1.$$

$$t = \frac{5.74}{\frac{9.69}{\sqrt{8}}} = 1.675$$

At significance level α , reject H_0 if $|t| > \underbrace{t_{n-1}(\alpha)}_{\text{c.v.}}$

$$\text{df} = n - 1 \Rightarrow \text{c.v.} = 2.365$$

Conclude there is sufficient statistical evidence of a difference in mean outcomes between X and Y . $\alpha = 0.05$



Stationary

A time series Y_t is defined to be covariance *stationary* if

- Focus
1. $E(Y_t) = \mu$ does not depend on t
 2. $\text{var}(Y_t) = \sigma^2$ does not depend on t

(later) 3. $\text{cov}(Y_t, Y_{t-j}) = \gamma_j$ depends only on j but not on t , for $j = 1, 2, \dots$

Asset Prices vs Returns

P_t
Asset prices (e.g. msft and sp500) do not seem to be stationary as they do not have a constant mean. These assets prices do not have the tendency to return to their means.

R_t r_t
Simple returns or log returns do seem to be stationary as they hover around some constant average/mean value over the sample period. Unlike asset prices, return has no obvious change in mean over the sample period.

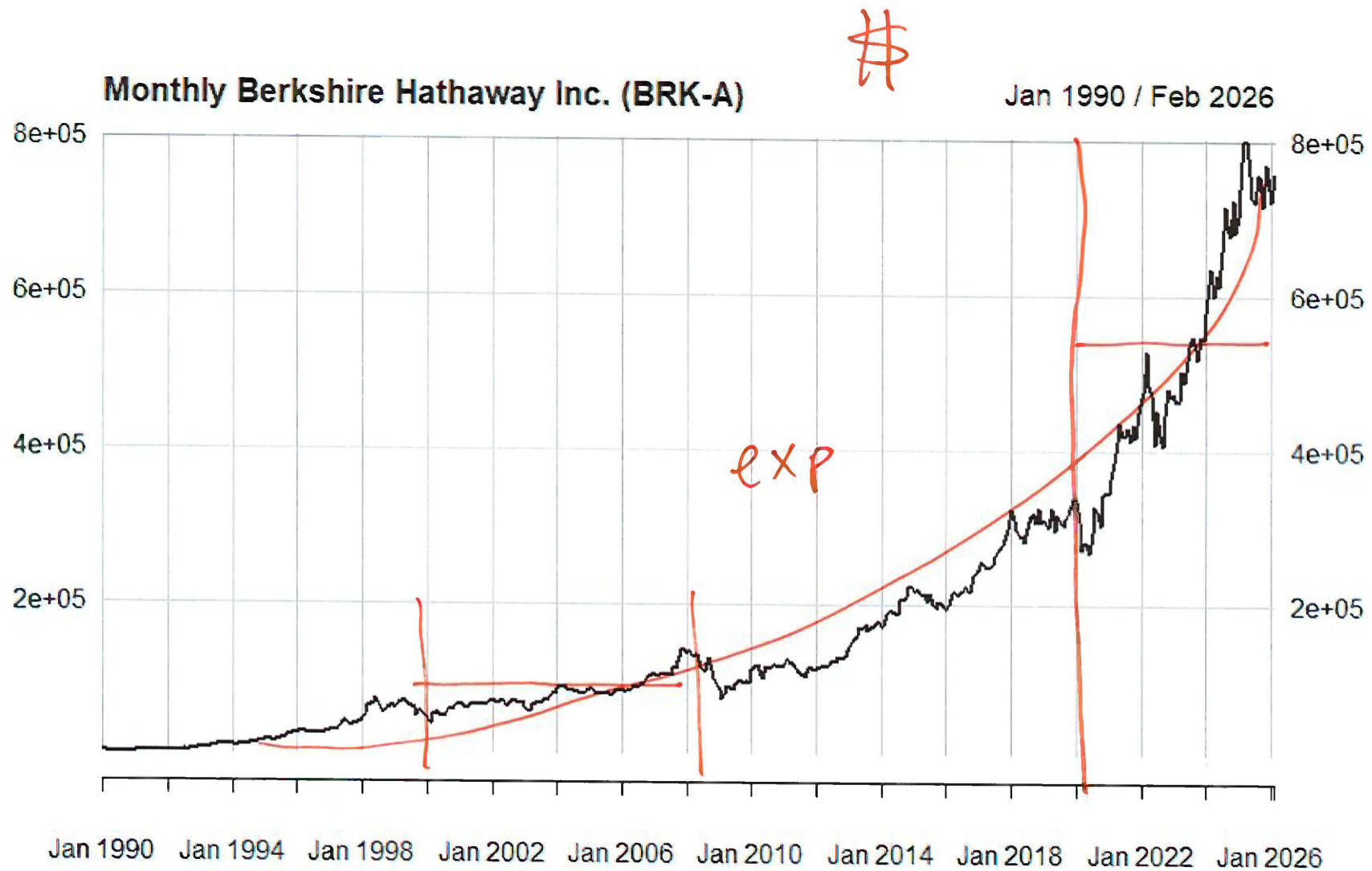
Berkshire Hathaway, BRK-A

```
> head(BRKA.M.Prices)
```

	BRKA
Jan 1990	7455
Feb 1990	7225
Mar 1990	6725
Apr 1990	6700
May 1990	7150
Jun 1990	7200

```
> tail(BRKA.M.Prices)
```

	BRKA
Sep 2025	754200
Oct 2025	715740
Nov 2025	770100
Dec 2025	754800
Jan 2026	722500
Feb 2026	757000



mean is not constant

Not Stationary.

Never report sample mean for #

Monthly Simple Returns

P_t

```
> BRKA.M.s.ret <- BRKA.M.Prices / lag(BRKA.M.Prices, 1) - 1  
> head(BRKA.M.s.ret)
```

P_{t-1}

```
BRKA  
Jan 1990 NA  
Feb 1990 -0.030851777  
Mar 1990 -0.069204152  
Apr 1990 -0.003717472  
May 1990 0.067164179  
Jun 1990 0.006993007
```

```
> tail(BRKA.M.s.ret)  
BRKA  
Sep 2025 -0.001429933  
Oct 2025 -0.050994431  
Nov 2025 0.075949367  
Dec 2025 -0.019867550  
Jan 2026 -0.042792793  
Feb 2026 0.047750865
```


Monthly Log Returns

```
> BRKA.M.1.ret <- diff(log(BRKA.M.Prices))
```

```
> head(BRKA.M.1.ret)
```

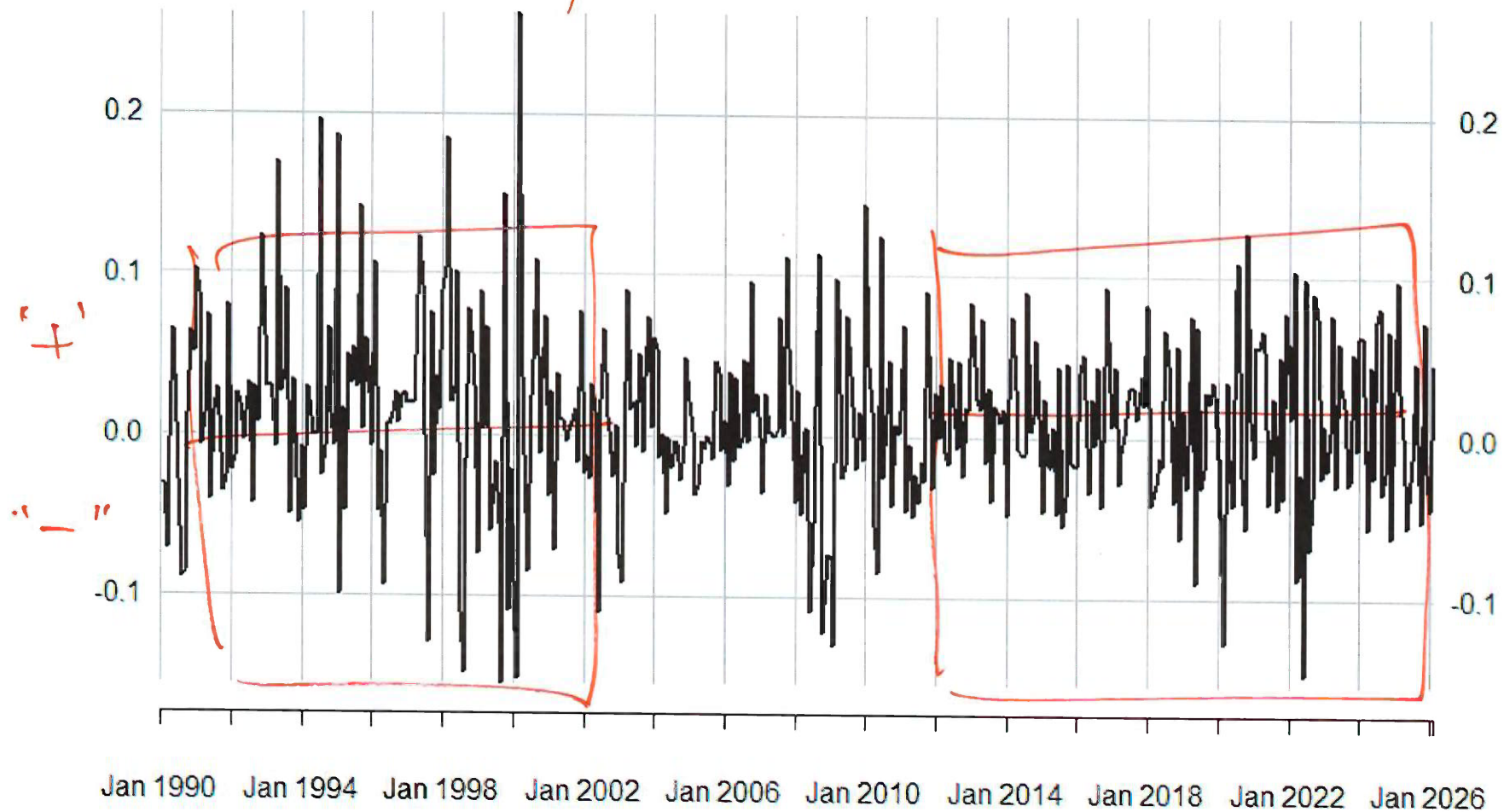
	BRKA
Jan 1990	NA
Feb 1990	-0.031337714
Mar 1990	-0.071715309
Apr 1990	-0.003724399
May 1990	0.065004830
Jun 1990	0.006968669

```
> tail(BRKA.M.1.ret)
```

	BRKA
Sep 2025	-0.001430957
Oct 2025	-0.052340612
Nov 2025	0.073203404
Dec 2025	-0.020067563
Jan 2026	-0.043735394
Feb 2026	0.046645833

Monthly Log returns BRK-A

Jan 1990 / Feb 2026



r_t or R_t stationary

make sense to calculate sample mean

Compute the 5% monthly Value-at-Risk (VaR) based on an initial \$1,000,000 investment.

```
> BRKA.M.1.ret <- na.omit(BRKA.M.1.ret)
> xbar.brka = mean(BRKA.M.1.ret)
> xbar.brka
[1] 0.01067085  $\bar{x} \Rightarrow \mu$ 
> sd.brka = sd(BRKA.M.1.ret)
> sd.brka
[1] 0.05574727  $\hat{\sigma}_{\text{sample}} \Rightarrow \sigma_{\text{pop}}$ 
> q.brka = xbar.brka + sd.brka *  $\overbrace{\text{qnorm}(0.05)}^{-1.645}$ 
> VaR.brka = q.brka * 1000000
> VaR.brka
[1] -81025.25  $\bar{x} + \sigma z_{0.05}$ 
```

With 5% probability, we could lose \$81025.25 or more over the next month out of our initial \$1,000,000 investment.

Downloading financial data from Yahoo Finance

```
install.packages("quantmod")
library("quantmod")

options("getSymbols.warning4.0"=FALSE)

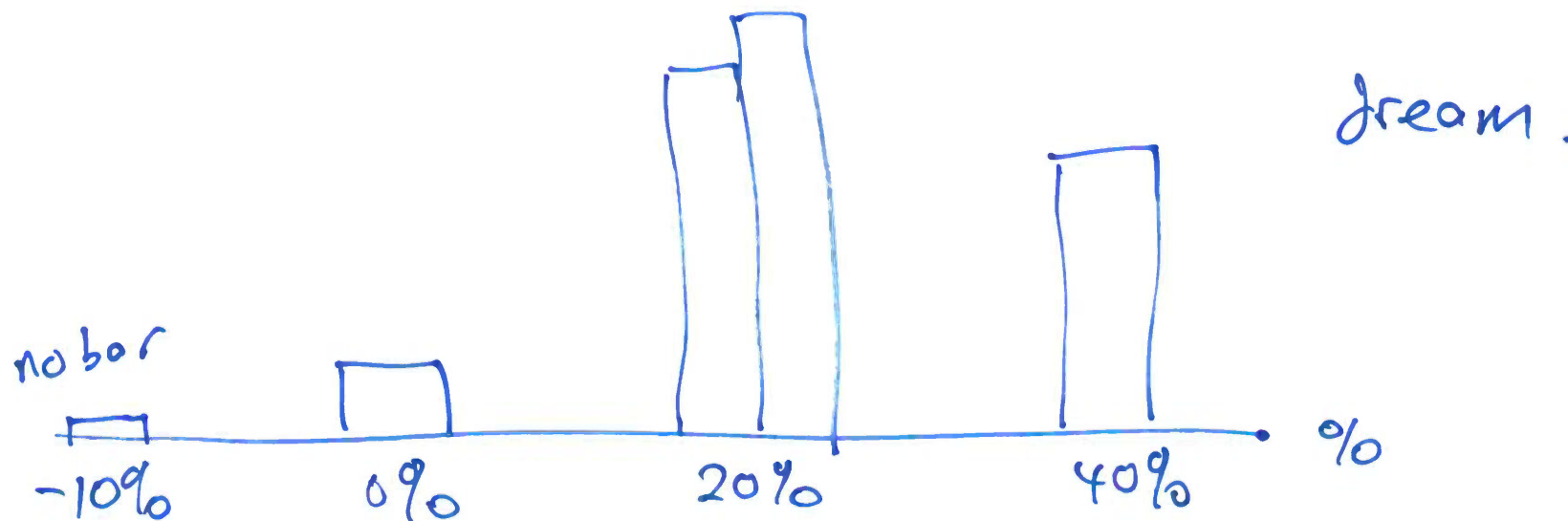
# ^GSPC is SP500
# BRK-A is Berkshire Hathaway
getSymbols(Symbols = c("BRK-A", "MSFT", '^GSPC'),
           from="1990-01-31", to="2020-12-31",
           auto.assign=TRUE, warnings=FALSE)
```

Histogram

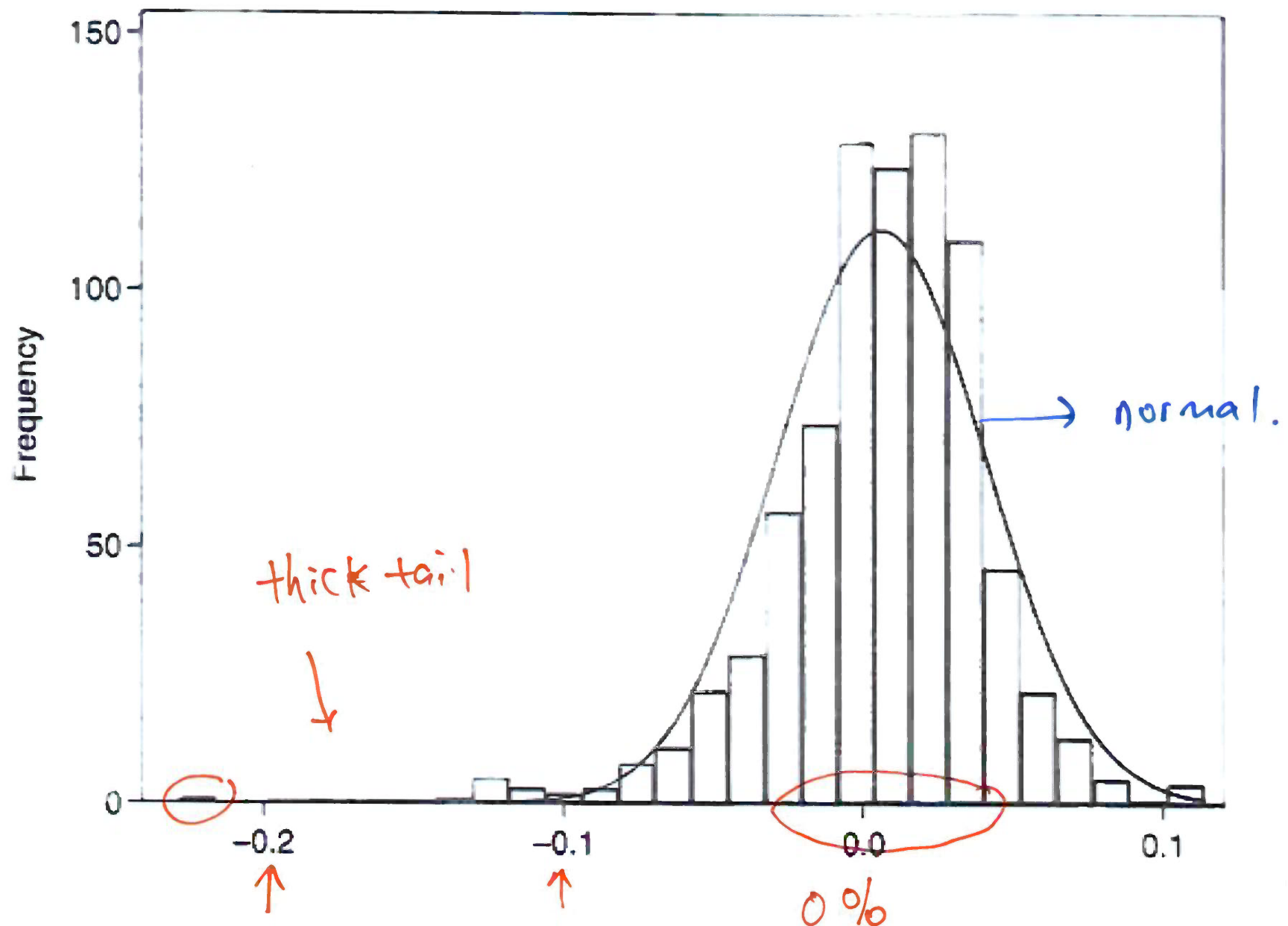
A histogram of returns is used to visualise the distribution, central tendency, and spread of a dataset.

Order returns from smallest to largest.

Each bin is represented by a bar, and the height of the bar indicates the number or frequency of data points that fall within that interval.



Histogram for the monthly log returns on the S&P 500. Superimposed on the returns distribution is the best fitting normal distribution.



Normality of Returns?

1. Recall that normal random variables have

- ▶ Symmetry $\implies$ **negative** returns are as likely as **positive** returns
- ▶ Thin tails $\implies$ likelihood of extreme outcomes is small.

2. Our data:

- ▶ Asymmetry $\implies$ **negative** returns are **more** likely than **positive** returns.
- ▶ Thick tails $\implies$ likelihood of extreme outcomes is large.

Features of asset returns imply non-normality. Large **negative** returns are much more likely than large **positive** returns.

Course Outline

1. This course will introduce you to the models and statistical methods commonly used in finance.
2. Just like the example, we will do this through analysis of existing stylized facts of asset returns.
3. This requires us to revise the concepts of:
 - ▶ random variables
 - ▶ expectations (means, variances, covariances)
 - ▶ conditional expectations

Week 2: Statistical Properties of Financial Data

Financial Assets

- ▶ Financial assets do not have physical worth (they are mostly pieces of paper). They derive their financial value from the fact that they represent a contractual claim on a stream of cash flows.
- ▶ Equities/stocks: give the owner an equity stake in a company (a share) and a claim on company's assets and earnings. These claims can be bought and sold on stock markets.
- ▶ Stocks give the owner the right to a payment which represents the distribution of some of the company's earnings, which is known as a dividend.
- ▶ The dividend is usually expressed as the amount each share receives or as a percentage of the current market price, referred to as dividend yield.

Stock Certificate

Figure Sapir Consolidated Airlines Stock



Equity Prices

- ▶ The price of an equity security is defined as the amount at which a transaction can occur (quoted price) or has occurred (historical price).
- ▶ Recording a “price” for the purposes of doing econometric analysis is not entirely straightforward. Multiple ways to define price:
 1. day’s closing price
 2. day’s opening price
 3. some markets even have overnight prices
- ▶ The adjusted closing price is generally used as the price because it accounts for stock splits and dividends. Therefore, we will generally use the adjusted closing price.

What does price tell us?

1. The goal of investing is to earn a profit. Price doesn't equal profit.
2. The most meaningful measure of profit for investors is return.
3. Return measures the appreciation of an asset.
4. It is measured as some difference between the current value of the asset, i.e., the current price, and its initial value, i.e., the price you paid for the asset.

Returns

- ▶ A financial return is a measure of the profit from investing in a financial asset.
- ▶ It accounts for the capital gain or loss due to the price change over the holding period of the asset.
- ▶ Also, it is impacted by the stream of cash flows (dividends) that occur over the course of the holding period.
- ▶ Data on prices are usually recorded at daily, weekly or monthly frequency, so holding period of the investment is limited to a multiple of this frequency.
- ▶ An investor could theoretically hold a financial asset forever, assuming the company never goes bankrupt.

Returns

Let P_t be the price of an asset at time t .

1. The simple **net** return over the period from $t - 1$ to t is

$$\textcircled{1} \quad R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1$$

2. The simple **gross** return is

$$\frac{P_t}{P_{t-1}} = 1 + R_t$$

$$\textcircled{1} \quad P_t = P_{t-1}(1 + R_t) \Rightarrow \text{slide 24}$$

$$\textcircled{2} \quad \frac{P_{t-1}}{P_{t-2}} = 1 + R_{t-1} \Rightarrow \text{slide 9} \quad \text{"t" with "t-1"}$$

Consider a one-month investment in Microsoft stock. Suppose you buy the stock in month $t - 1$ (e.g., end of February) at $P_{t-1} = \$85$ and sell the stock the next month (e.g., end of March) for $P_t = \$90$.

Jan $\leftarrow P_{t-2} = 80$
 Feb $P_{t-1} = 85$
 March $P_t = 90$

The one-month simple net and gross returns are then:

$$\text{net } R_t = \frac{\$90 - \$85}{\$85} = \frac{\$90}{\$85} - 1 = 1.0588 - 1 = 0.0588$$

$$\text{gross } 1 + R_t = 1.0588$$

The one-month investment in Microsoft yielded a 5.88% per month return.

\$10

Alternatively, \$1 invested in Microsoft stock in month $t - 1$ grew to \$1.0588 in month t .

\$10.58 $\leftarrow$
 $\downarrow$
 gross $\Rightarrow$ related equity curve.

Multi-period returns

1:00
1:05 back

The simple two-month return on an investment in an asset between months $t - 2$ and t is defined as:

$$R_t (2) = \frac{P_t - P_{t-2}}{P_{t-2}} = \frac{P_t}{P_{t-2}} - 1. \quad \Rightarrow \frac{\text{March} - \text{Jan}}{\text{Jan}}$$

Writing

$$\frac{P_t}{P_{t-2}} = \frac{P_t}{P_{t-1}} \frac{P_{t-1}}{P_{t-2}}$$

the two-month return can be expressed as:

net

$$\begin{aligned} R_t (2) &= \frac{P_t}{P_{t-1}} \frac{P_{t-1}}{P_{t-2}} - 1 \\ &= \underbrace{(1 + R_t)}_{\substack{\text{btw} \\ \text{march} \\ + \\ \text{Feb}}} (1 + \underbrace{R_{t-1}}_{\substack{\text{btw} \\ \text{Feb} \\ + \\ \text{Jan}}}) - 1. \end{aligned}$$

slide 7.

Multi-period returns

Then the simple two-month **gross** return becomes:

$$1 + R_t(2) = (1 + R_t)(1 + R_{t-1}) = 1 + R_{t-1} + R_t + \underbrace{R_{t-1}R_t}_{\text{cross product}}$$

Hence, the two-month **net** return is

$$R_t(2) = R_{t-1} + R_t + \underbrace{R_{t-1}R_t}_{\text{annoying}}$$

If, however, R_{t-1} and R_t are close to zero then $R_{t-1}R_t \approx 0$ and $1 + R_t(2) \approx 1 + R_{t-1} + R_t$ so that

$$R_t(2) = R_{t-1} + R_t. \quad \text{add}$$

Multi-period returns

Continuing with the previous example, suppose that the price of Microsoft stock in month $t - 2$ is \$80.

The two-month net return is:

$$\textcircled{1} \quad R_t \textcircled{2} = \frac{\overset{\text{march}}{\$90} - \overset{\text{Jan}}{\$80}}{\underset{\text{Jan}}{\$80}} = \frac{\$90}{\$80} - 1 = 1.1250 - 1 = 0.1250$$

or 12.50% per two-month.

Multi-period returns

② The two one-month returns are:

$$R_{t-1} = \frac{\overset{F}{\$85} - \overset{I}{\$80}}{\$80} = 1.0625 - 1 = 0.0625$$

$$R_t = \frac{\overset{M}{\$90} - \overset{F}{\$85}}{\$85} = 1.0588 - 1 = 0.0588$$

The two-month gross return

$$1 + R_t(2) = (1.0588)(1.0625) = 1.1250$$

$(1 + R_t)(1 + R_{t-1})$ ↗

Returns

In general, return of holding an asset for k periods, $R_t(k)$, is then

$$\begin{aligned} R_t(k) &= \frac{P_t}{P_{t-k}} - 1 \\ &= \frac{P_t}{P_{t-1}} \times \frac{P_{t-1}}{P_{t-2}} \times \dots \times \frac{P_{t-k+2}}{P_{t-k+1}} \times \frac{P_{t-k+1}}{P_{t-k}} - 1 \\ &= (1 + R_t) \times (1 + R_{t-1}) \times \dots \times (1 + R_{t-k+2}) \times (1 + R_{t-k+1}) - 1 \\ &= \prod_{j=0}^{k-1} (1 + R_{t-j}) - 1 \end{aligned}$$


equity curve

Log returns

The **log return** is denoted by r_t and is defined as

$$r_t = \log(1 + R_t) = \log\left(\frac{P_t}{P_{t-1}}\right) = \log(P_t) - \log(P_{t-1}).$$

Log returns are approximately equal to net returns because if x is small, then $\log(1 + x) \approx x$.

R_t very small

$$r_t = \log(1 + R_t) \approx R_t$$

Multi-period returns

Consider the two-month log return defined as:

$$\begin{aligned} r_t(2) &= \log(P_t) - \log(P_{t-2}) \\ &= [\log(P_t) - \log(P_{t-1})] + [\log(P_{t-1}) - \log(P_{t-2})] \\ \text{nice!} \quad &= r_t + r_{t-1} \quad \Rightarrow \text{add up} \Rightarrow \text{No cross product} \\ &\quad \downarrow \quad \downarrow \\ &\quad \text{M + Feb} \quad \text{Feb + Jan} \end{aligned}$$

Hence the two-month log return is just the sum of the two one-month log returns.

slide 10

$$R_t(2) = R_t + R_{t-1} + \text{annoying } R_t R_{t-1}$$

Multi-period returns

The k -period log return is

$$\begin{aligned} r_t(k) &= \log P_t - \log P_{t-k} \\ &= (\log P_t - \log P_{t-1}) + (\log P_{t-1} - \log P_{t-2}) + \cdots \\ &\quad + (\log P_{t-k+1} - \log P_{t-k}) \\ &= r_t + r_{t-1} + \cdots + r_{t-(k-1)} \\ &= \sum_{j=0}^{k-1} r_{t-j} \end{aligned}$$

$k=2$ $\sum_{j=0}^1 r_{t-j} = r_t + r_{t-1}$

The k -period log return is simply the sum of the single period log returns over the pertinent period.

Compute two-month log returns

Using the price and return data from the previous example, the log monthly return on Microsoft stock can be computed in two ways:

or

$$r_t = \ln(1.0588) = 0.0571$$

$\downarrow$
 $1 + R_t \Rightarrow \text{slide 8}$

$\Rightarrow$ given R_t
OR

$$r_t = \ln(90) - \ln(85) = 4.4998 - 4.4427 = 0.0571$$

$\downarrow$
 P_t
 m
 $\downarrow$
 P_{t-1}
Feb

give P_t
 P_{t-1}

Compute two-month log returns

Using the data from the previous example, the two-month log return on Microsoft stock can be computed in two equivalent ways.

The first way uses the difference in the logs of P_t and P_{t-2}

$$\textcircled{1} \quad r_t(2) = \ln(90) - \ln(80) = 4.4998 - 4.3820 = 0.1178.$$

P_t P_{t-2}
 m Jan

Compute two-month log returns

The second way uses the sum of the two one-month log returns. Here

$$r_t = \ln(\overset{P_t}{90}) - \ln(\overset{P_{t-1}}{85}) = 0.0571$$

M *F* +

$$r_{t-1} = \ln(\overset{P_{t-1}}{85}) - \ln(\overset{P_{t-2}}{80}) = 0.0607$$

Feb *Jan*

so that

$$r_t(2) = 0.0571 + 0.0607 = 0.1178$$

↓

Dividends

What about dividends?

In the Microsoft example, we assume that no dividend is paid between months $t - 1$ and t .

For multi-period returns, we again assume that no dividend is paid between months $t - 2$ and t .

Dividends

Adjusting the calculation of returns for the payment of dividends, denoted by D_t , between time $t - 1$ and t is relatively straightforward.

For simple net and gross returns

$$R_t = \frac{P_t + \overset{\text{new}}{D_t} - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} + \overset{\text{new}}{\frac{D_t}{P_{t-1}}} - 1$$

$$1 + R_t = \frac{P_t + D_t}{P_{t-1}} = \frac{P_t}{P_{t-1}} + \frac{D_t}{P_{t-1}}$$

Relative price (P_t/P_{t-1}) and the dividend yield (D_t/P_{t-1}).

Dividends

- ▶ For log returns

$$r_t = \log(1 + R_t) = \log\left(\frac{P_t + D_t}{P_{t-1}}\right) = \log\left(\frac{P_t}{P_{t-1}} + \frac{D_t}{P_{t-1}}\right)$$

- ▶ Much of the unit will ignore the issue of dividends.
- ▶ This stems from the reality that dividends are paid relatively infrequently.
- ▶ Also they constitute a minor proportion of the overall return compared with price movements.

Portfolio Returns

Often concerned not with the return to a single asset but rather the return to a **portfolio** of financial assets.

Consider a portfolio with only two assets whose portfolio shares are w_1 and w_2 respectively, with the normalization

$$w_1 + w_2 = 1.$$

0.5 + 0.5 equal weight

slide 7 gross return

$$P_t = w_1 P_{t-1} (1 + R_{1,t}) + w_2 P_{t-1} (1 + R_{2,t})$$

↓ value \$ portfolio. ↓ A1 ↓ A2

Using simple gross returns for each asset, the value of the portfolio between $t - 1$ and t may be calculated as

$$\begin{aligned} \$ &= P_t = w_1 P_{t-1} (1 + R_{1t}) + w_2 P_{t-1} (1 + R_{2t}) \leftarrow \\ &= P_{t-1} (w_1 (1 + R_{1t}) + w_2 (1 + R_{2t})) \end{aligned}$$

We can rearrange as

$$(1 + R_{pt}) \equiv \frac{P_t}{P_{t-1}} = \underbrace{w_1 (1 + R_{1t})} + \underbrace{w_2 (1 + R_{2t})}$$

Expanding yields

$$1 + R_{pt} = \underbrace{w_1 + w_2}^1 + w_1 R_{1t} + w_2 R_{2t}$$

so that

$$\begin{array}{c} \text{return} \rightarrow \\ \text{for} \\ \text{port.} \end{array} R_{pt} = w_1 R_{1t} + w_2 R_{2t}.$$

Portfolio Returns

In simple returns, the portfolio return is just the weighted sum of individual returns.

$$R_{pt} = w_1 R_{1t} + w_2 R_{2t}.$$

This is true for N returns

$$R_{pt} = \sum_{i=1}^N w_i R_{it},$$

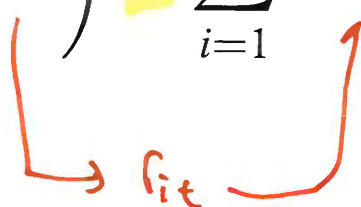
any N

with

$$\sum_{i=1}^N w_i = 1.$$

Portfolio Returns: Log Returns

This result does not extend to the case of log returns

$$r_{pt} = \log(1 + R_{pt}) = \log\left(1 + \sum_{i=1}^N w_i R_{it}\right) \neq \sum_{i=1}^N w_i r_{it}.$$


How to combine log returns into the portfolio return?

Portfolio Returns: Log Returns

Recall

$$r_t = \log(\underbrace{1 + R_t}) = \log\left(\underbrace{\frac{P_t}{P_{t-1}}}\right).$$

Take the exponential of both sides to give:

$$e^{r_t} = 1 + R_t = \frac{P_t}{P_{t-1}}$$

Rearranging we get

$$\underline{P_t} = P_{t-1} e^{r_t}$$

annoying.

exp not r_t

$$\exp(\ln(x)) = x$$

Portfolio Returns: Log Returns

The value of the portfolio between $t - 1$ and t may be calculated as

$$P_t = \underbrace{w_1 P_{t-1} e^{r_{1t}}}_{A1} + \underbrace{w_2 P_{t-1} e^{r_{2t}}}_{A2}$$

so that

$$r_{pt} \equiv \log \left(\frac{P_t}{P_{t-1}} \right) = \log (w_1 e^{r_{1t}} + w_2 e^{r_{2t}})$$

For N assets,

$$\text{return} \Rightarrow r_{pt} = \log \left(\sum_{i=1}^N w_i e^{r_{it}} \right)$$

not $\sum_{i=1}^N w_i r_{it} \Rightarrow$ I want this !!!

$\downarrow$
 R_{it}

Portfolio Returns: Log Returns

- ▶ Most of the time we **simply ignore** this formula

$$r_{pt} = \log \left(\sum_{i=1}^N w_i e^{r_{it}} \right)$$

- ▶ Instead we define

$$r_{pt} := \sum_{i=1}^N w_i r_{it}.$$

$N=2$

$= w_1 r_{1,t} + w_2 r_{2,t}$

- ▶ We simply take a weighted aggregate of these log returns to find portfolio returns.

Stock Indices

- ▶ An aggregate summary measure of the performance of the stock market as a whole is known as a stock market index.
- ▶ Indices combine a selection of (a large number of) stocks in a particular way to create a portfolio.
- ▶ Stock market indices are price indices. The computation of returns to the index can be performed in exactly the same way as if it were a single stock.

Stock Indices: Two common ways of construction: the market.

1. Price-weighted index (PWI) is a stock market index in which each stock is given a weight based on its price per share. This means that stocks with a higher share price have a larger effect on the index.

2. Value-weighted index (VWI) is a stock market index in which each stock is given a weight based on its total market value, usually measured by market capitalisation.

Company	Price		# of shares		Total value.
A	\$ 100	x	1	=	\$ 100
B	\$ 10	x	100	=	\$ 1,000

Handwritten notes: An arrow points from "SP 500" to the table. In the "Total value" column, an arrow points from "\$ 1,000" to "\$ 100" with the text "bigger size".

PWI $\Rightarrow$ A matters more $\$100 > \$10 \Rightarrow$ more weights on higher P_t

VWI $\Rightarrow$ B matters more $\$1000 > \$100 \Rightarrow$ " " or bigger size.

Stock Indices

- ▶ Deutscher Aktien Index (DAX): comprises the 30 largest German companies that trade on the Frankfurt Stock Exchange (VWI).
- ▶ Dow Jones Industrial Average (DJ or DJIA) Index: computed using 30 prominent United State corporations (PWI).
- ▶ Financial Times Stock Exchange 100 (FTSE) Index: 100 largest companies listed on the London Stock Exchange (VWI).
- ▶ Hang Seng (HSX) Index: 40 of the largest companies that trade on the Hong Kong Exchange (VWI).

Stock Indices

- ▶ **Nikkei 225** (SKX) Index: 225 prominent companies listed on the Tokyo Stock Exchange (**PWI**).
- ▶ Standard and Poors Composite 500 (S&P500 or S&P) index: 500 large companies having common stock listed on the NYSE, NASDAQ, or the Cboe BZX Exchange. It is computed by summing the market value of the outstanding equity in each firm in the index (**VWI**).

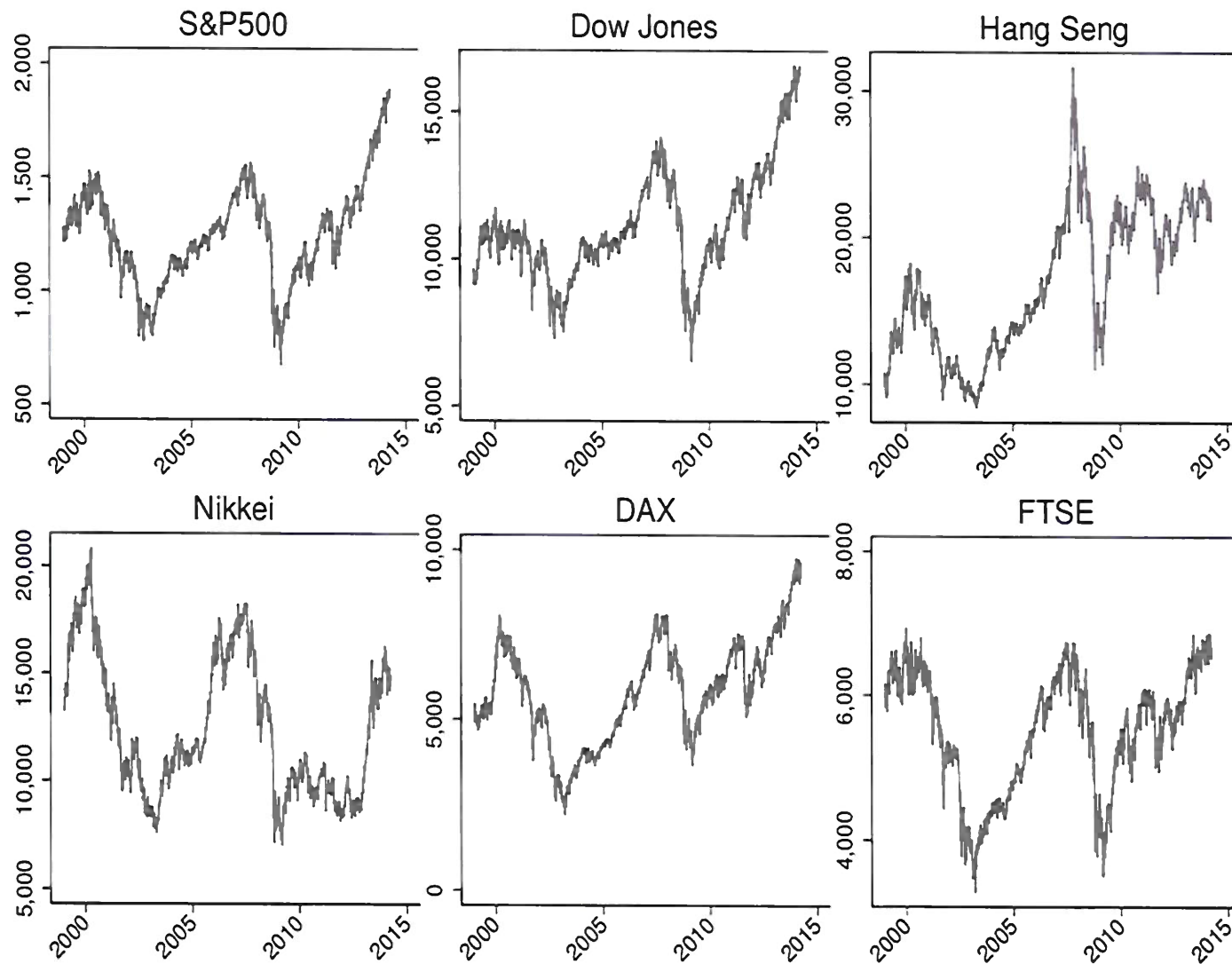


Figure 1.2: Daily observations on six international stock market indices for the period 4 January 1999 to 2 April 2014.

What Do Investors Care About

Intuitively, there are at least four factors an investor would like to know when considering investing in a risky asset.

1. The expected return from investing in the asset.
2. The uncertainty surrounding the value of, or payoff from, the investment in the asset. In other words, the chance that the actual return on an investment may be very different than the expected return.
risk $\Rightarrow$ std dev $\Rightarrow$ volatility.
3. Whether or not the extreme returns are above or below the expected value.
4. The relative likelihood of occurrence of extreme returns (positive or negative).

What tools do we need?

To explain these specific behaviors, we must first properly define them. We will revisit some concepts of probability and statistics.

Recall:

- ▶ the different types of random variables: continuous and discrete
- ▶ the “moments”, or expectations, of different random variables, and how they are estimated from sample data

Random Variables

$$P_t = \$90$$

$$P_{t+1} = \$110 \Rightarrow \text{R.V.}$$

Random Variables are numbers that we are not sure about but we might have some idea of how to describe its potential outcomes.

Consider the price of Microsoft stock next month, P_{t+1} . Since P_{t+1} is not known with certainty today (t), we can consider it as a random variable.

The two most common classes of random variables are called:

1. Discrete
2. Continuous

Discrete Random Variables

$$0 \leq p \leq 1$$

↓
what is p ?

Definition

A discrete random variable X is one that can take on a finite number of n different values $S_X = x_1, x_2, \dots, x_n$.

Definition

The probability mass function (**pmf**) of a discrete random variable, denoted $p(x)$, is a function such that

$$p(x) = \Pr(X = x)$$

↖ R.V.

↘ outcome

e.g.

$$p(1) = \Pr(X = 1) = p \Rightarrow X=1 \text{ success.}$$

$$p(0) = \Pr(X = 0) = 1 - p \quad X=0 \text{ otherwise failure}$$

← Bernoulli

$$\begin{aligned} \text{e.g. } X &= 1 && \text{if } r_k > 0 \\ X &= 0 && \text{otherwise.} \end{aligned}$$

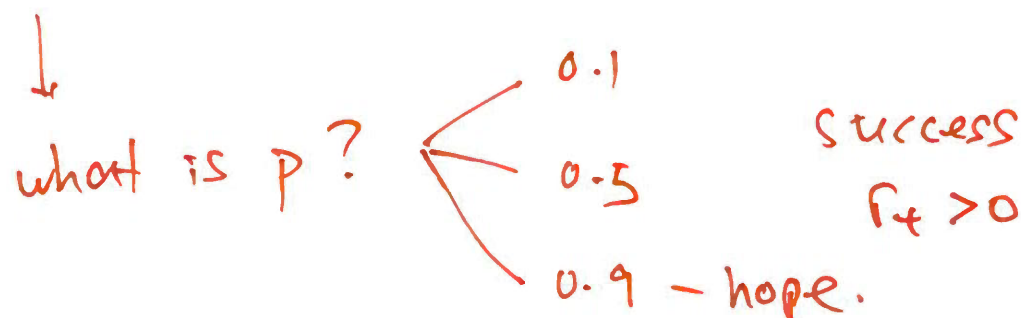
Bernoulli

A Bernoulli random variable takes values $X = 1$ with probability p and $X = 0$ with probability $1 - p$.

Bernoulli random variables are often used to model “success” or “failure”, where success is loosely defined.

For example, a **success** could be a stock price going up, a coin landing on heads, a bull market or a corporate default and a **failure** could be a stock price going down, a coin landing on tails, a bear market or a corporate solvency.

Often, denoted as $X \sim \text{Bern}(p)$.



Example

Let r_t be the log return on the S&P500 index at time t .

Let $X = 1$ if $r_t > 0$ and $X = 0$ else.

$X \sim \text{Bern}(p)$

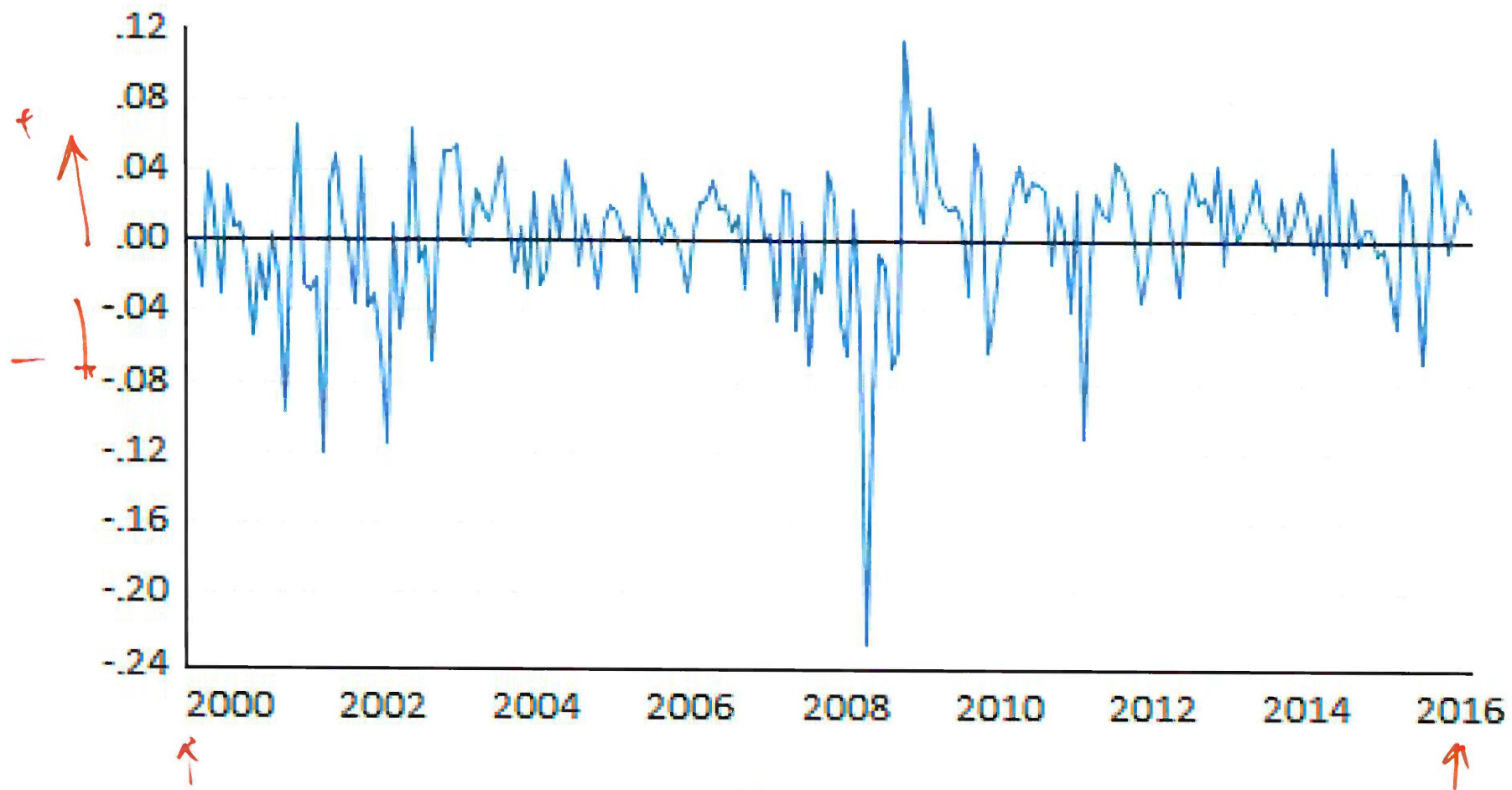
How would you estimate p ?

Mathematically, for n denoting the total number of observations, we can estimate p by

$$\text{hat } p \Rightarrow \hat{p} = \frac{\#\{r_t > 0\}}{n}$$

$\#\{r_t > 0\}$ is the count of positive returns.

S&P 500 monthly log returns, January 2000 to August 2016



We have $n = 200$, $\#\{r_t > 0\} = 121$ and hence $\hat{p} = 0.605 = \frac{121}{200}$

200 returns $\begin{cases} r_t > 0 \\ \text{or} \\ r_t < 0 \end{cases}$

Continuous Random Variables

Definition

A continuous random variable X is one that can take on any real value. That is, $S_X = x : x \in \mathbb{R}$

Definition

not pmf $\Rightarrow$ discrete.

The probability density function (**pdf**) of a continuous random variable X is a nonnegative function f , defined on the real line, such that for any interval A :

$$\Pr(X \in A) = \int_A f(x) dx$$

$\rightarrow$ p.d.f $\Rightarrow$ formula.

That is, $\Pr(X \in A)$ is the area under the probability curve over the interval A .

not prob $\Rightarrow$ $f(x) = \frac{1}{\sqrt{2\pi}} \exp^{-\frac{1}{2}x^2} \Rightarrow$ give me x

Continuous Random Variables: Normal

The most famous continuous random variable is the “normal random variable”. Often, denoted as $X \sim N(\mu, \sigma^2)$.

A normal random variable X has pdf

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}.$$

μ is the location parameter; σ is the scale parameter.

The standard normal distribution

Special case is the “standard normal”, $\mu = 0$ and $\sigma = 1$.

If a random variable X follows a standard normal distribution then we often write $X \sim N(0, 1)$ as short-hand notation.

The pdf of a standard normal random variable is given by:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$$

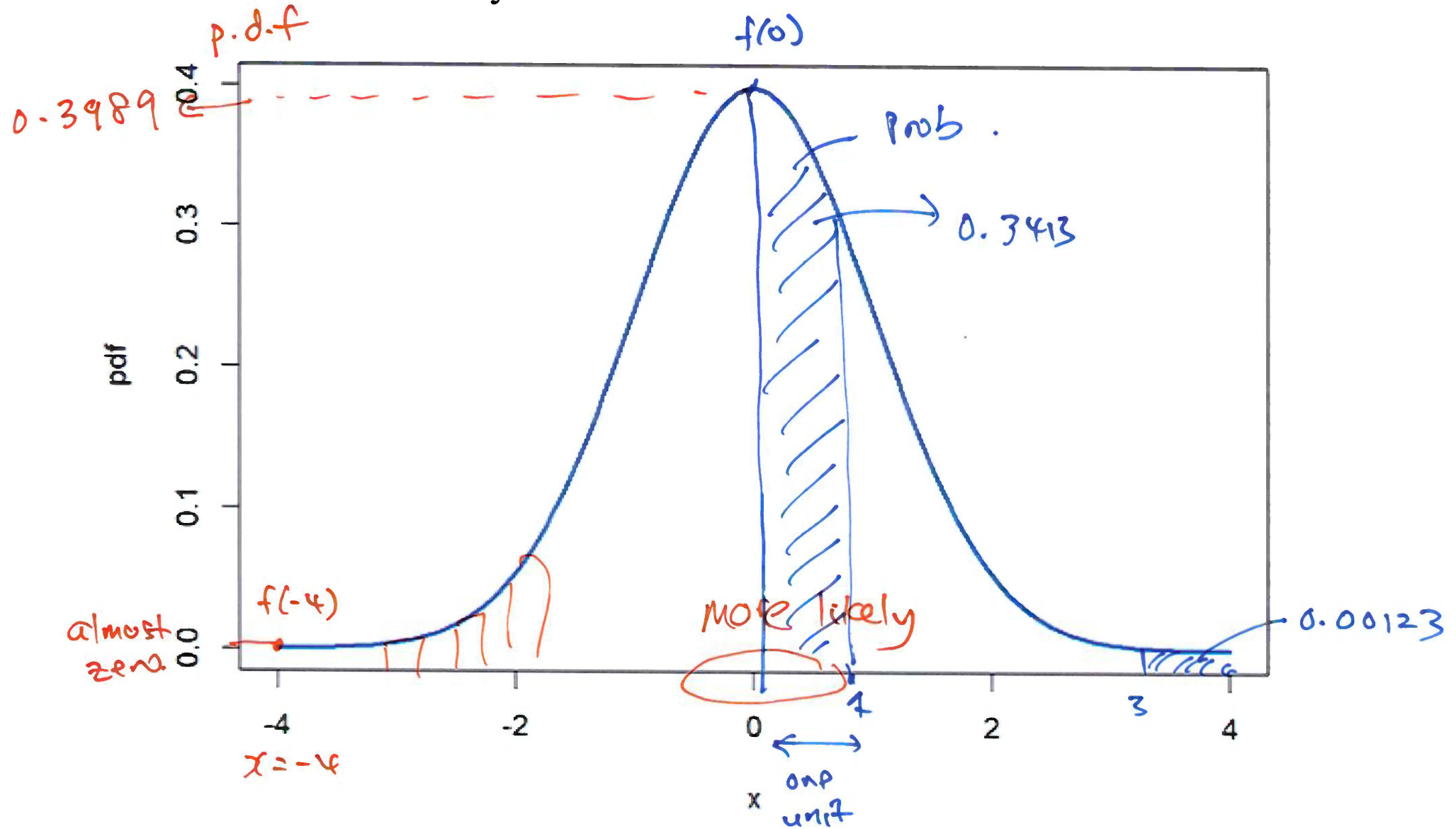
with $-\infty \leq x \leq \infty$.

$x = 0 \quad f(0) = 0.3989 \dots$

$x = -4 \quad f(-4) = \text{almost zero}$

like "pub."

Standard normal density



Normal distribution

The normal distribution has the annoying feature that the area under the normal curve cannot be evaluated analytically.

That is, $\Pr(X \in A)$ is the area under the probability curve over the interval A :

$$\Pr(a \leq X \leq b) = \int_a^b \underbrace{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}}_{f(x)} dx$$

does not have a closed form solution.

Areas under the normal curve, in one form or another, are given in tables in every introductory statistics book and standard statistical software can be used to find these areas.

$\Pr(-2 \leq X \leq 1)$ is given by the area under the probability curve.

